Chapter 1

First Look: Ten Questions.

1. Put options with strikes 30 and 20 on the same underlying asset and with the same maturity are trading for \$6 and \$4, respectively. Can you find an arbitrage?

2. The number 2^{29} has 9 digits, all different. Without computing 2^{29}, find the missing digit.

3. Let W_t be a Wiener process, and let

$$X_t = \int_0^t W_\tau d\tau.$$

What is the distribution of X_t? Is X_t a martingale?

4. Alice and Bob stand at opposite ends of a straight line segment. Bob sends 50 ants towards Alice, one after another. Alice sends 20 ants towards Bob. All ants travel along the straight line segment. Whenever two ants collide, they simply bounce back and start traveling in the opposite direction. How many

ants reach Bob and how many ants reach Alice? How many ant collisions take place?

5. Find all the values of ρ such that

$$\begin{pmatrix} 1 & 0.6 & -0.3 \\ 0.6 & 1 & \rho \\ -0.3 & \rho & 1 \end{pmatrix}$$

is a correlation matrix.

6. How many independent random variables uniformly distributed on $[0, 1]$ should you generate to ensure that there is at least one between 0.70 and 0.72 with probability 95%?

7. Show that the probability density function of the standard normal integrates to 1.

8. Assume the Earth is perfectly spherical and you are standing somewhere on its surface. You travel exactly 1 mile south, then 1 mile east, then 1 mile north. Surprisingly, you find yourself back at the starting point. If you are not at the North Pole, where can you possibly be?!

9. Solve the Ornstein-Uhlenbeck SDE

$$dr_t = \lambda(\theta - r_t)dt + \sigma dW_t,$$

which is used, e.g., in the Vasicek model for interest rates.

10. Write a C++ function that computes the n-th Fibonacci number.